Mithun Vanniasinghe

647-782-0797 mithun.vanniasinghe@mail.utoronto.ca linkedin.com/in/mithun-vanniasinghe mithunvvv.github.io

Research Interests

Robotics | Control Theory | Safe Control | Reachability | Reinforcement Learning | Imitation Learning

Education

National University of Singapore **Aug 2026 – Apr 2030**
PhD in Electrical and Computer Engineering *Singapore*

University of Toronto **Sep 2021 – Apr 2026**
Bachelor of Applied Science in Engineering Science *Toronto, Canada*

- **Specialization:** Major in Machine Intelligence, Minor in Robotics
- **Academic Standing:** High Honours, cGPA: 3.85/4.0, Dean's Honour List (2021–2026)
- **Relevant Coursework:**

ECE356 – Introduction to Control Theory (A+)	ECE421 – Machine Learning (A+)
ECE324 – Machine Intel., Software & Neural Nets (A+)	ROB311 – Artificial Intelligence (A)
ECE367 – Matrix Algebra & Optimization (A)	ECE368 – Probabilistic Reasoning (A)
ECE557 – Linear Control Theory (IPR)	ECE411 – Adaptive Control & Reinforcement Learning (IPR)
ROB501 – Computer Vision for Robotics (IPR)	ECE470 – Robot Modeling & Control (IPR)

Research Experience

Undergrad Thesis Student **Sep/25 – Apr/26**
LEAF Lab (Advisors: Nick Rhinehart) *University of Toronto*

- Investigating Hamilton-Jacobi reachability analysis and bisimulation-based representation learning to enhance safety in robotic systems, with the goal of improving both sample efficiency and operational safety.

Imitation Learning & Robotics Researcher **May/25 – Sep/25**
Autonomous Space Robotics Lab (Advisors: Tim Barfoot, Nick Rhinehart) *University of Toronto*

- Developed and implemented an imitation learning model, Ratatouille, for human-robot social navigation. **Submitted as second author to ICRA 2026.**
- Designed and conducted real-world experiments, debugging hardware and managing distributed computation across devices using ROS communication.
- Debugged multimodal dynamics issues and identified key factors required for successful behavioral cloning, enabling robust model tuning and real-world deployment in human-robot social navigation.

Reinforcement Learning Researcher **Aug/24 – Sep/25**
DORL Lab (Advisors: Jiayu Chen, Chi-Guhn Lee) *University of Toronto, University of Hong Kong*

- Co-developed LOKI, a hierarchical imitation learning framework for offline skill discovery, introducing a novel extrinsic-intrinsic objective (this idea served as the basis for the paper.) **Submitted as co-first author to ICRA 2026.**
- Built the codebase for the initial phase of the alignment-enforced Vector Quantized VAE and implemented a novel skill-segmentation pipeline.
- Conducted all experiments and performed detailed time-series analysis of learned skills, including the collection, preprocessing, and cleaning of the offline trajectory dataset to ensure high-quality inputs for hierarchical imitation learning.

Machine Learning Researcher **May/23 – Present**
Super CDMS International Collaboration (Advisors: Scott Oser, Yan Liu) *University of British Columbia*

- Led the implementation of time series GANs, adapting model architectures for specific use cases to enhance blinding schemes. Utilized Python and TensorFlow to generate synthetic time series sensor data, reducing bias in experimental analyses.
- Applied ML techniques, including PCA and t-SNE, to validate the effects of dimensionality reduction in embedding spaces during time series GAN training.

Physics Researcher **May/22 – Aug/22**

Super CDMS International Collaboration (Advisors: Miriam Diamond) *University of Toronto*

- Designed Python testing protocols, utilizing the unittest framework, to decrease the risk of signal analysis software failure across a wide range of machines and hundreds of users. Leveraged GitLab CI/CD and Singularity/Apptainer containerization.
- Conducted benchmarking tests via Bash scripts in High Performance Computing environments and developed Python programs for runtime analysis informing software engineers about computing needs of the collaboration.

Publications

Ratatouille: Imitation Learning Ingredients for Real-world Social Robot Navigation

Submitted to ICRA 2026

- J. R. Han, M. Vanniasinghe, H. Sahak, N. Rhinehart, and T. D. Barfoot, "Ratatouille: Imitation Learning Ingredients for Real-world Social Robot Navigation," *arXiv preprint* arXiv:2509.17204, 2025. [Video Demo]

Offline Discovery of Interpretable Skills from Multi-Task Trajectories

Submitted to ICRA 2026

- C. Zhu, M. Vanniasinghe, J. Chen, and C.-G. Lee, "Offline Discovery of Interpretable Skills from Multi-Task Trajectories, 2025.

Teaching Experience

Linear Algebra Teaching Assistant **Sep/23 – Present**

Linear Algebra for Engineers *University of Toronto*

Calculus Teaching Assistant **Sep/23 – Present**

Calculus for Engineers *University of Toronto*

Industry Experience

Applied Machine Learning Engineer **May/24 – Present**

Tenstorrent Inc. *Toronto, Canada*

- **Model Bring-up + Inference Server:** Deployed Mistral 7B on internal inference server, integrated with ML model studio platform, and resolved complex batching and context leakage issues.
- **Model Benchmarking + Infrastructure:** Benchmarked 18+ CV, NLP, and diffusion models on custom hardware, modified 50+ demo scripts for multi-batch and dual-chip support, and added performance logging for internal metrics visibility.
- **LLM App Development + Evaluation:** Built real-time LLM agent app for internal demo day (search + multi-threaded chat), integrated Llama 3.1 70B for live inference, and implemented log-likelihood evaluation using vLLM for grounded model comparisons.

Technical Skills

Languages: Python, C, C++, Bash
Libraries: NumPy, TensorFlow, PyTorch, Keras, SciPy, Scikit-Learn, Matplotlib
Technologies/Frameworks: ROS2, MATLAB, Git, High Performance Computing, Docker

Awards and Honours

- Dean's Undergraduate Student Summer Research Pivot Fellowship $8000 **May/25**
- Gary L. Palmer Memorial Scholarship $1600 **Aug/23**
- Herbert Gladish Memorial Scholarship $1100 **Aug/23**
- Nat. Sciences and Eng. Research Council of Canada Undergraduate Student Research Award $10,500 **May/23**
- Engineering Science Research Opportunity Program Award $7500 **May/22**

Leadership and Mentorship

VP Technical Writing, UofT Machine Intelligence Student Team (UTMIST) **Jul/25 – Present**

- Coordinated monthly AI/ML newsletter article publications across a team of 15+ writers.
- Initiated and managed the conversion of written articles into engaging short-form video content for social media.

Engineering Mentor (NSight) **Sep/24 – Present**

- Provide 1:1 mentorship to first-year engineering students, offering guidance on academic success, internship and research experience, study skills, and transitioning to university.

Engineering Orientation Leader **Sep/22 – Present**

- Served as orientation group leader to welcome incoming first-year students; volunteered at welcoming ceremonies, Frosh Week, and the club fair.